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7. Worked examples

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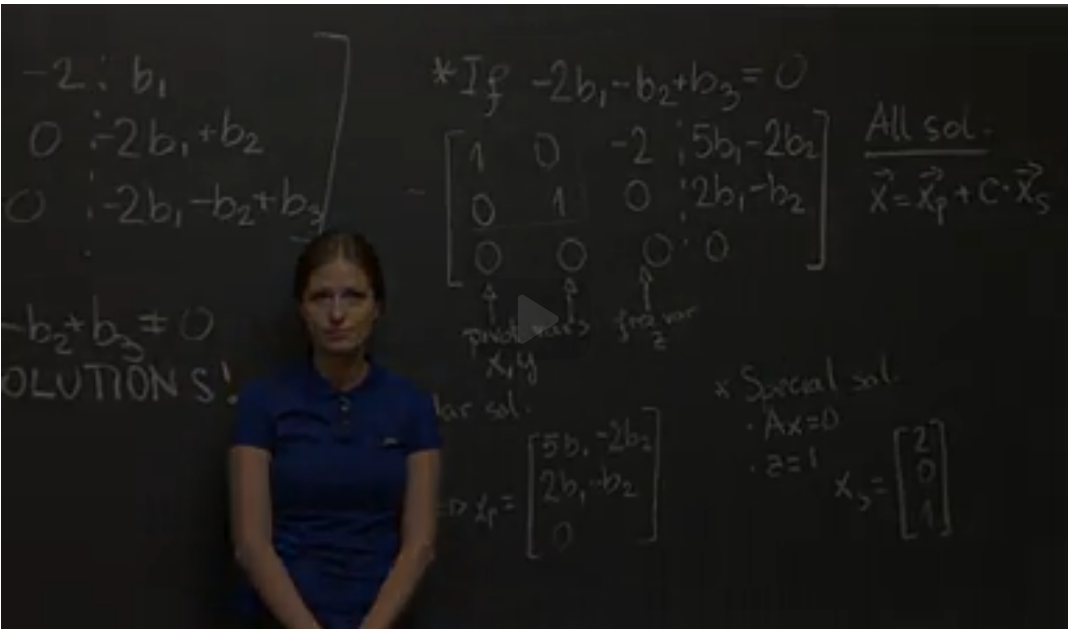
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Solving a matrix equation



In case this particular combination of parameters is not zero, there are no solutions.

In case this particular combination of parameters is equal to zero, there are solutions.

There are as many of them as there are real numbers C.

And they are all of this form for these two vectors.

And that's all I wanted to say today.

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Question 7.1

Choose a value of **b** for which the system is inconsistent, and another value of **b** for which the system is consistent and has infinitely many solutions. What are two of those solutions?

$$\begin{cases} 3x + 2y = 10 \\ 6x + 4y = b. \end{cases}$$

(3.1)

Worked Solution

We put the system into augmented matrix form:

$$\left(\begin{array}{cc|c} 3 & 2 & 10 \\ 6 & 4 & b \end{array} \right).$$

(3.2)

We can eliminate the second equation by subtracting **2** times the first equation from it, leaving us with

$$\left(\begin{array}{cc|c} 3 & 2 & 10 \\ 0 & 0 & b - 20 \end{array} \right).$$

(3.3)

The second row reads $0 \cdot x + 0 \cdot y = b - 20$. If $b \neq 20$, then the equation can never be satisfied, in which case, there would be no solutions to the system. If $b = 20$, then there are infinitely many solutions to the system. The first row tells us that

$$3x + 2y = 10,$$

(3.4)

or that

$$x + \frac{2}{3}y = \frac{10}{3}.$$

(3.5)

Setting $y = c_1$ we see that

$$x = \frac{10}{3} - \frac{2}{3}c_1,$$

(3.6)

so

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10/3 \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} -2/3 \\ 1 \end{pmatrix}.$$

(3.7)

You should check to see if the two solutions you found can be written in this form. To get any one solution plug in a specific number for c_1 .

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